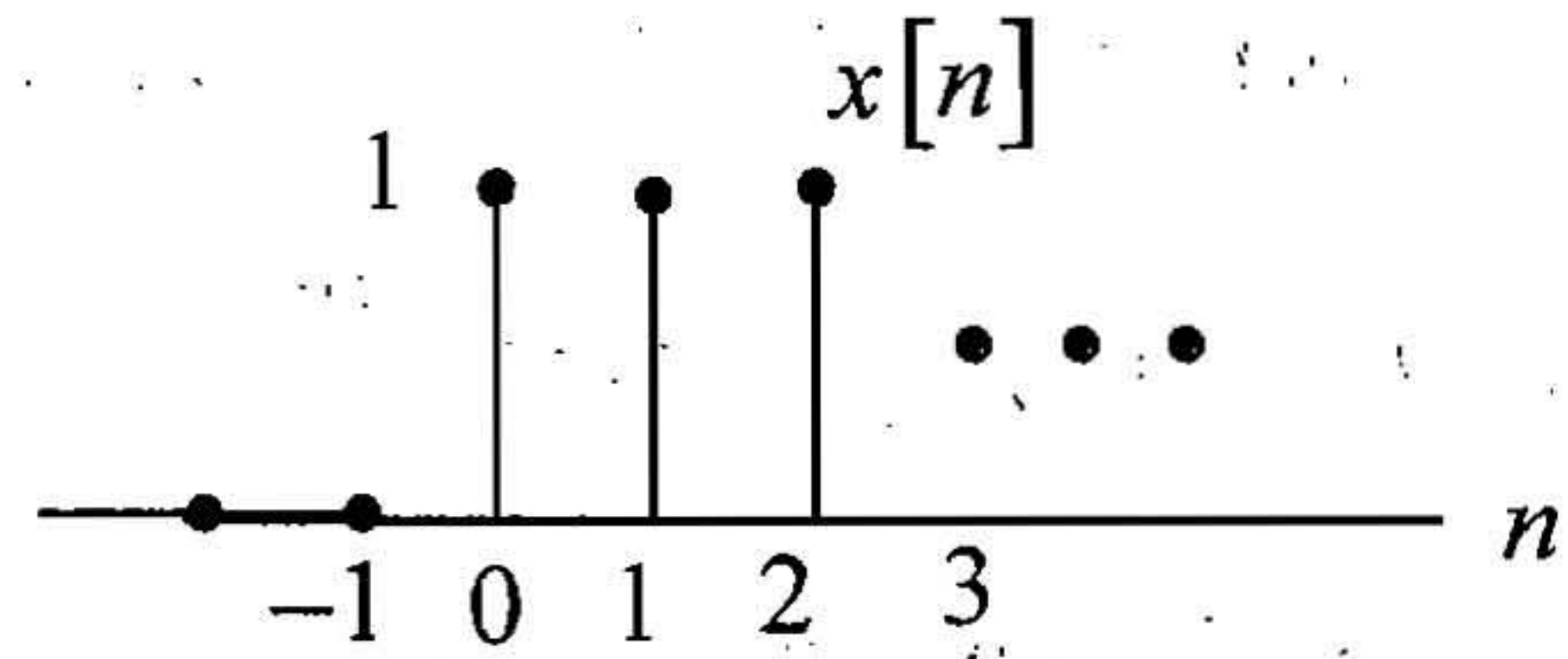


电子科技大学

2009 年攻读硕士学位研究生入学试题参考解答

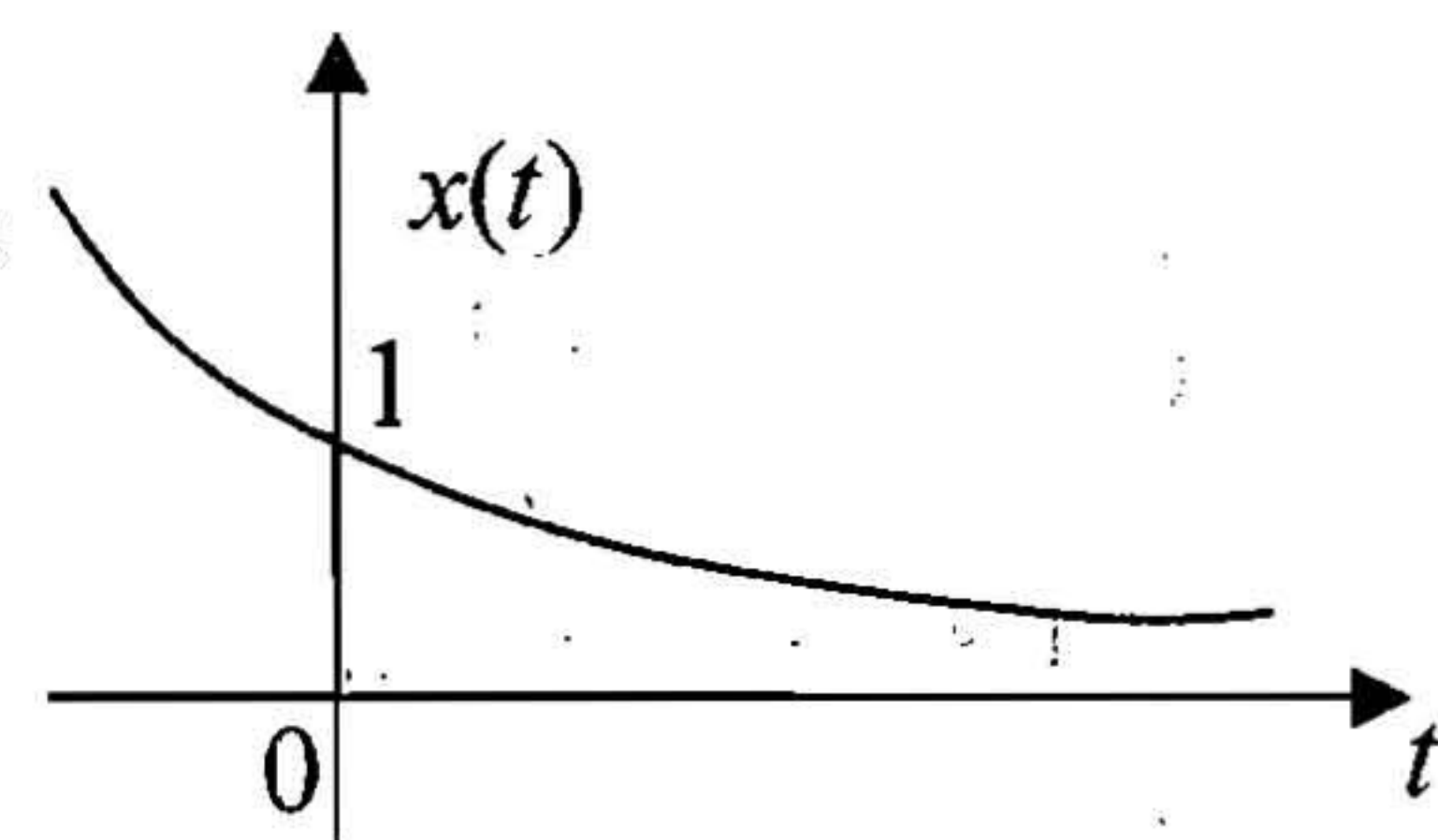
考试科目：836 信号与系统和数字电路

1. 解：(1) $x[n] = u[n]$ (4 分)



(3 分)

(2) $x(t) = e^{-t} \quad -\infty < t < +\infty$ (4 分)

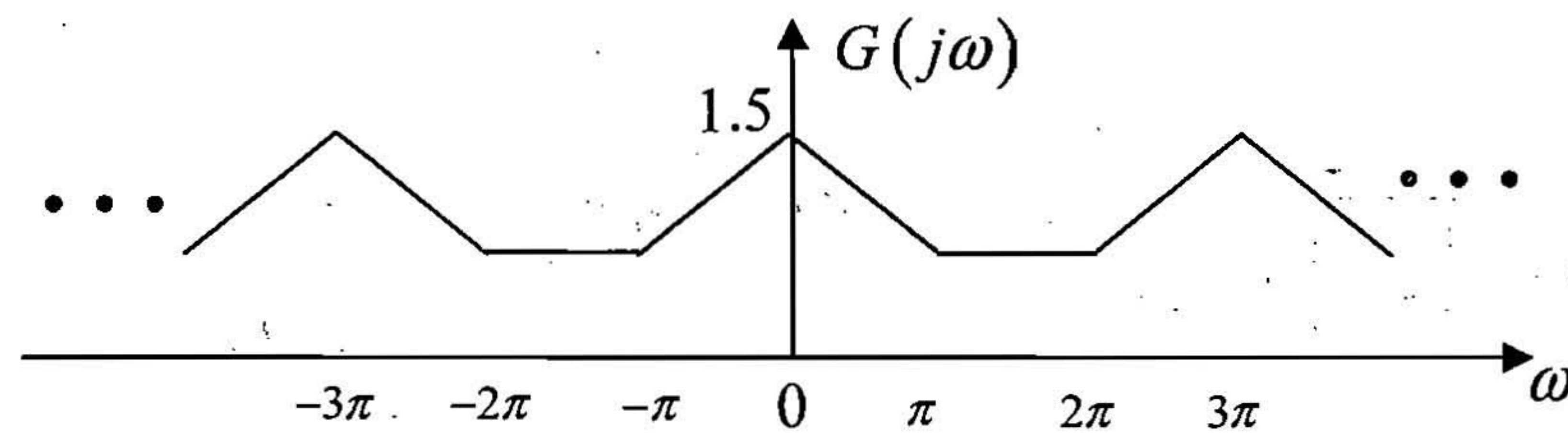


(4 分)

2. 解： $X(j\omega) = \begin{cases} X(j\omega) & |\omega| \leq 2\pi \\ 0 & \text{others} \end{cases}$ (3 分)

$$G(j\omega) = 1.5 \sum_{k=-\infty}^{+\infty} X[j(\omega - 3k\pi)] \quad (4 \text{ 分})$$

$\max\{\omega_0\} = \pi$ (4 分)



(4 分)

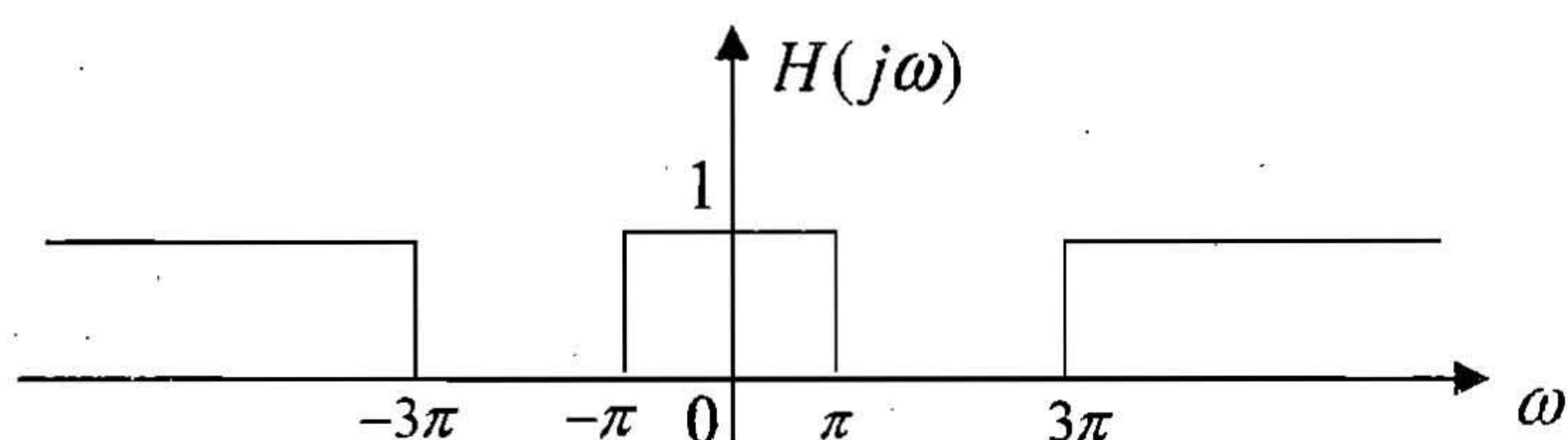
3. 解：(1) 没必要限制 (3 分)

(2) $|\omega_c| > 100\pi$ (6 分)

4.解:

$$(1) H(j\omega) = 1 - H_1(j\omega) + H_2(j\omega)$$

$$= \begin{cases} 1 & |\omega| < \pi \text{ or } |\omega| > 3\pi \\ 0 & \text{others} \end{cases}$$



$\therefore H(j\omega)$ 实现了理想带阻滤波器(4分)

$$h(t) = F^{-1}\{H(j\omega)\} = \delta(t) - \frac{\sin 3\pi t}{\pi t} + \frac{\sin \pi t}{\pi t} \quad (4 \text{ 分})$$

$$(2) y(t) = 1 + \sin 4\pi t \quad (7 \text{ 分})$$

5. 解:

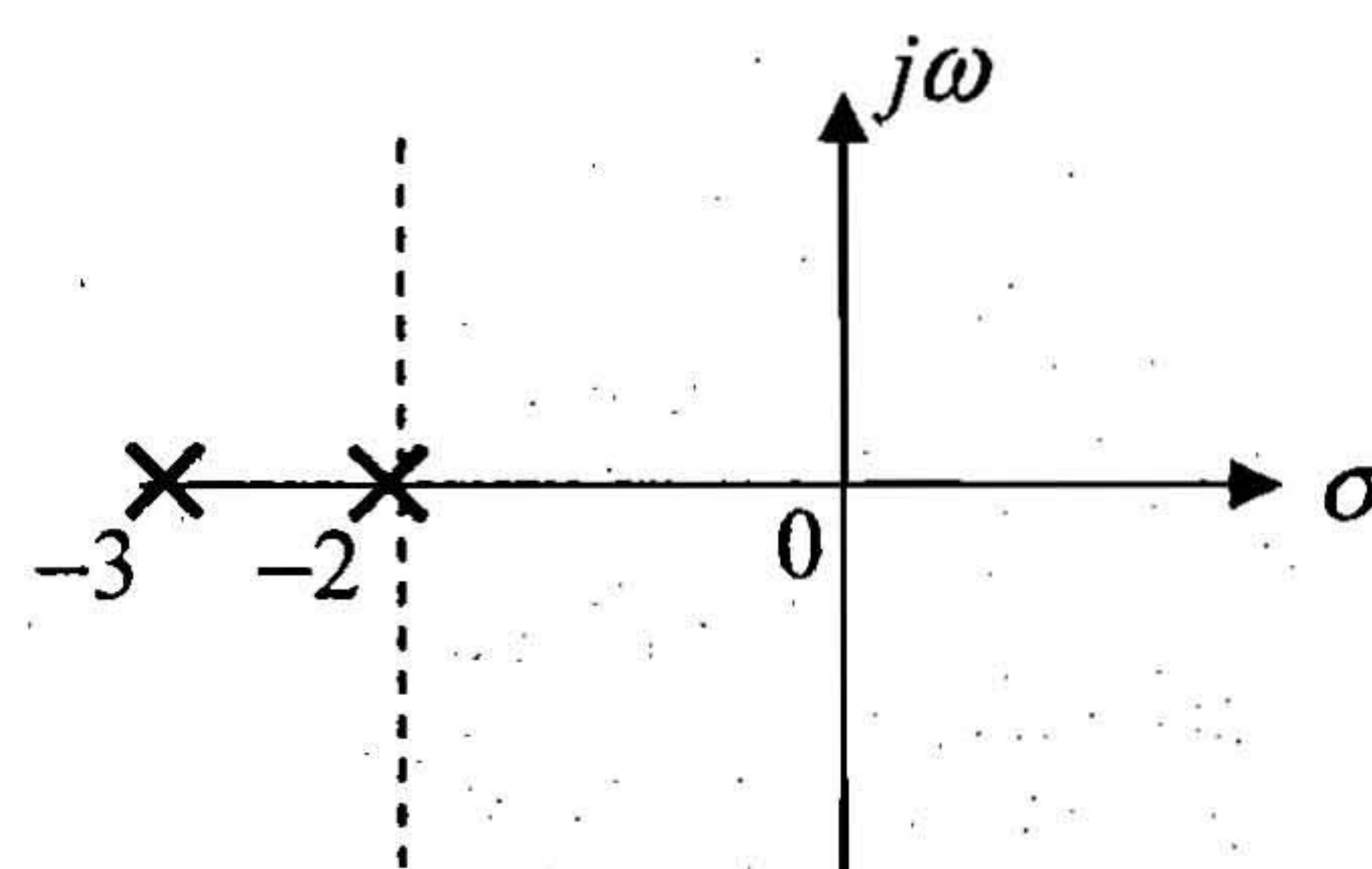
$$(1) H(s) = \frac{1}{(s+2)(s+3)} = \frac{1}{s^2 + 5s + 6}, \quad \text{ROC: } \text{Re}\{s\} > -2 \quad (3 \text{ 分})$$

一阶极点位于 $s = -2$ 和 $s = -3$ 处, 无零点。

$$(2) h(t) = (e^{-2t} - e^{-3t})u(t) \quad (3 \text{ 分})$$

该系统是稳定系统。(3分)

$$(3) y(t) = \left[\frac{1}{2}e^{-t} - e^{-2t} + \frac{1}{2}e^{-3t} \right] u(t) \quad (4 \text{ 分})$$



$H(s)$ 的零极点图 (2分)

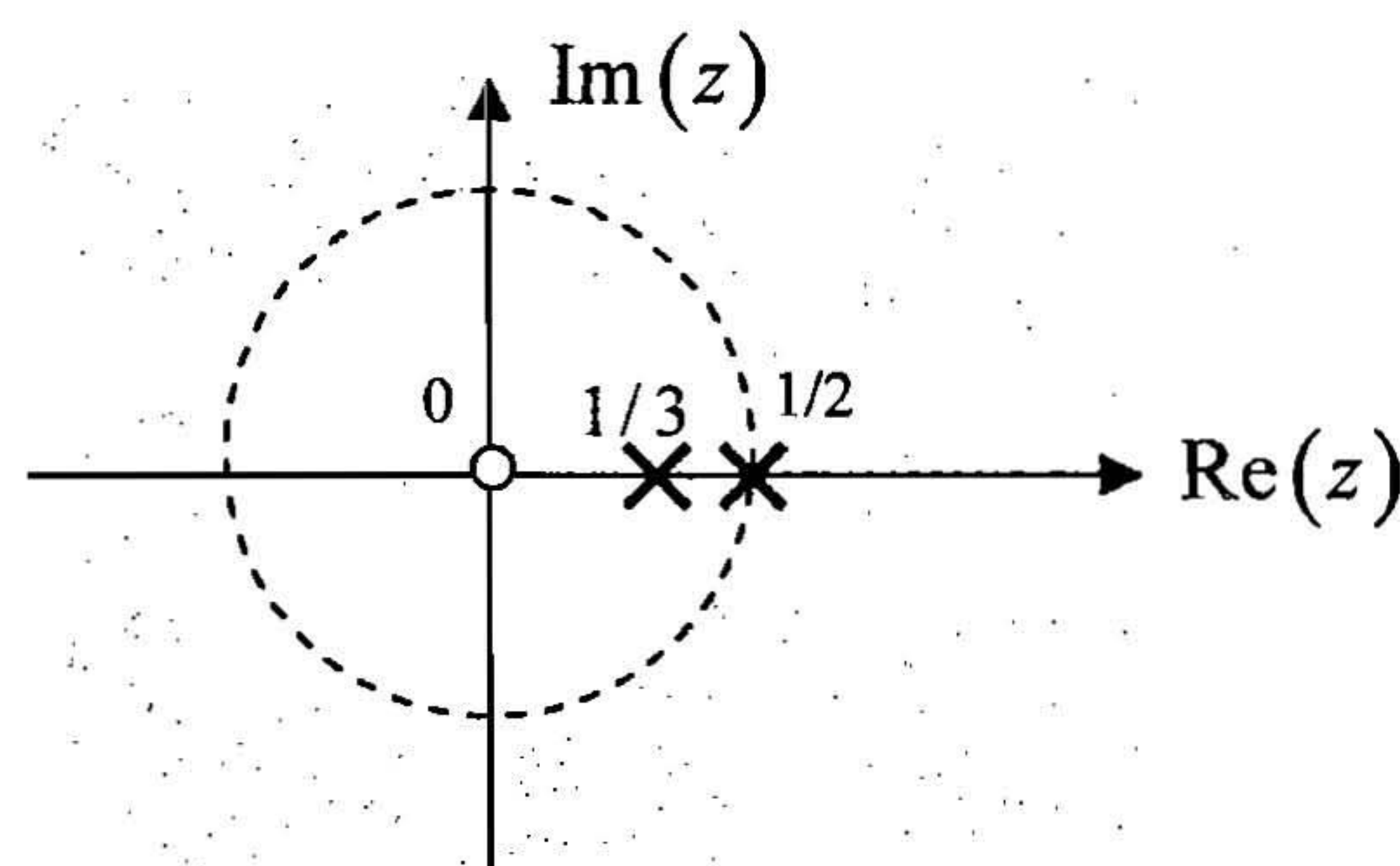
$$(4) \frac{d^2}{dt^2} y(t) + 5 \frac{d}{dt} y(t) + 6y(t) = x(t) \quad (3 \text{ 分})$$

6.解:

$$(1) H(z) = \frac{z^{-1}}{\left(1 - \frac{1}{3}z^{-1}\right)\left(1 - \frac{1}{2}z^{-1}\right)} \quad \text{ROC: } |z| > \frac{1}{2} \quad (4 \text{ 分})$$

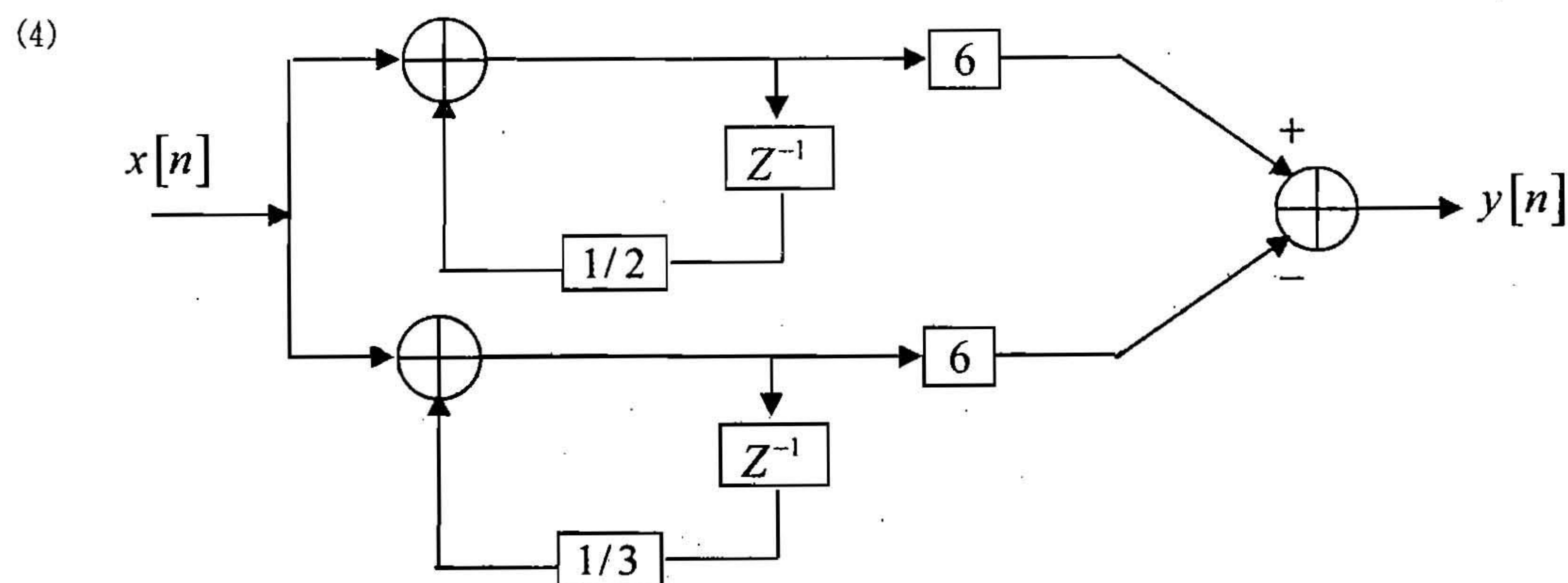
$$(2) h[n] = 6 \left[\left(\frac{1}{2}\right)^n - \left(\frac{1}{3}\right)^n \right] u[n] \quad (2 \text{ 分})$$

该系统是因果的。(2分)



(2分)

(3) $y[n] - \frac{5}{6}y[n-1] + \frac{1}{6}y[n-2] = x[n-1]$ (4 分)



(4 分)

7、a c c d b

8、10110100 10001101

01100010 100001

高 低

(0, 3, 5, 6) (1, 2, 4, 7)

3, 4

9、激励输入方程:

$$D_1 = X \cdot Q_2^n \cdot \overline{Q_1^n} + \overline{X} \cdot Q_1^n + \overline{Q_2^n} \cdot Q_1^n$$

$$D_2 = X \cdot \overline{Q_2^n} + Q_1^n \overline{Q_2^n} + \overline{X} \cdot \overline{Q_1^n} \cdot Q_2^n$$

输出方程式: $Z = X \oplus \overline{Q_1^n}$

波形图 (略)

10、

$Q_B Q_A$					
$Q_D Q_C$		$Q_D Q_C$			
		00	01	11	10
00		0	0	1	1
01		0	0	1	1
11		1	1	0	0
10		1	1	0	0

$$F = Q_B \overline{Q_D} + \overline{Q_B} Q_D = Q_B \oplus Q_D$$

用 74x151 实现上述函数存在多种方式（略）

74X194 电路连接中 S1S0=01